

Tire Model Fitting

From the raw TTC datasets, we have data for pure longitudinal and pure lateral for 18" tire but only lateral for 16" tires. But in the next iteration of our FS car, we wish to incorporate 16" tires as doing this will help us reduce mass, rotating inertia and doing this could have a significant effect on the lap times.

That's why it was very essential for us to have specific tire models for evaluation of longitudinal as well as lateral forces for both 18" and 16" tires.

We use the Pacejka Magic Formula 5.2 to fit the tire model from the processed data.

Pre-Processing

We carefully study and observe the raw TTC datasets and based on that, try removing the noise and make the dataset uniform and smooth.

- We first round off the F_z values to the nearest 100, apply a median filter to remove noise and then classify all the values of F_z into 5 different classes namely -1100N, -900N, -600N, -400N, -200N.
- Then for pressure of the tire, we deploy the same classification approach and classify all the values of pressure into 4 different classes namely 56KPa, 70KPa, 84KPa, 98KPa.
- The same thing has been replicated for the camber inclination of the tire as well by classifying the camber values into 4 distinct classes namely -1deg, 0deg, 1deg, 2deg.

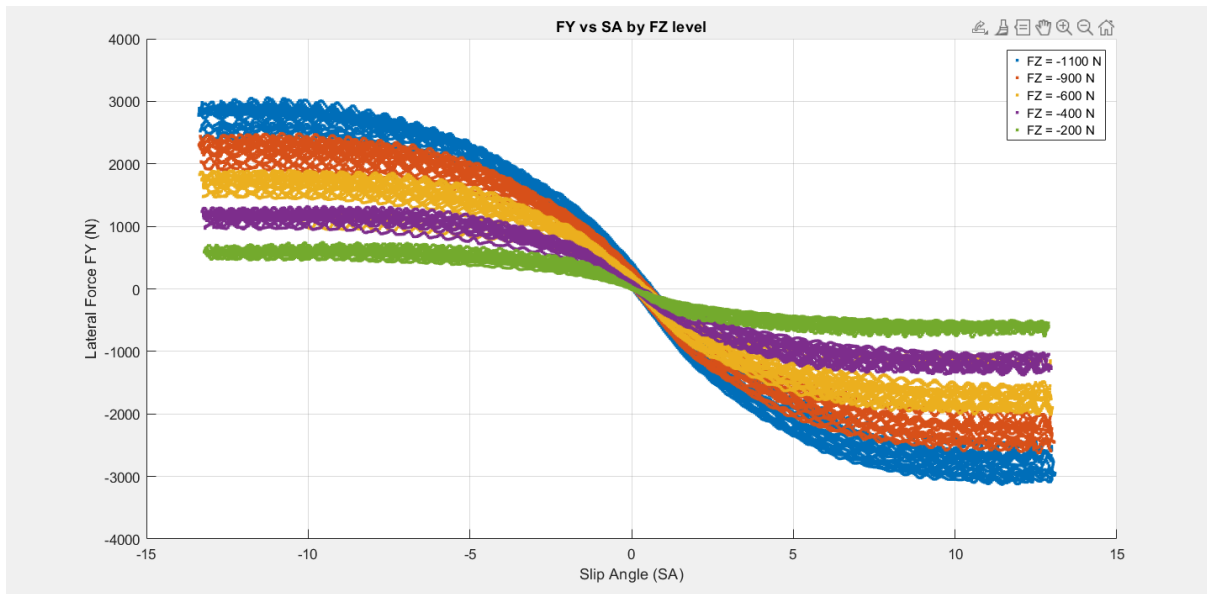
After doing these 3 pre-processing steps to each of the three datasets we have, we obtain fairly uniform and smoothened data and we plot each one of them before fitting the tire model (F_y v/s SA for cornering data and F_x v/s SR for longitudinal data)

Post Processing Plots

Pure Cornering 18"

Dataset used:

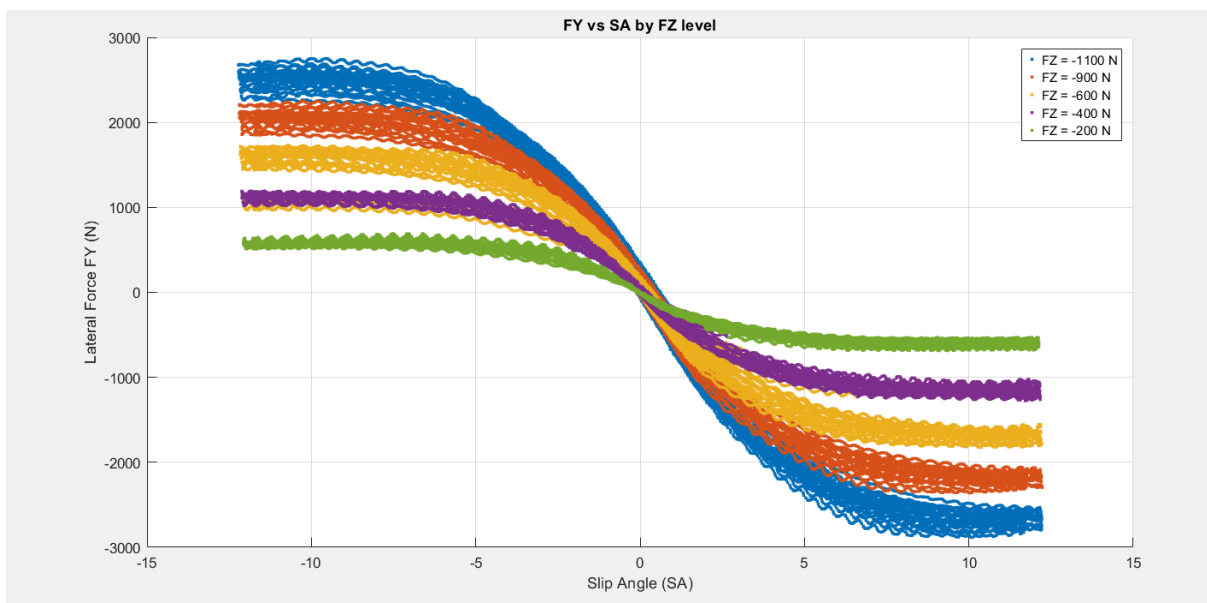
<https://drive.google.com/file/d/1nH7DTK3zGD1fHcb7hotFkjgw53Sm1A8u/view?usp=sharing>



Pure Cornering 16"

Dataset used:

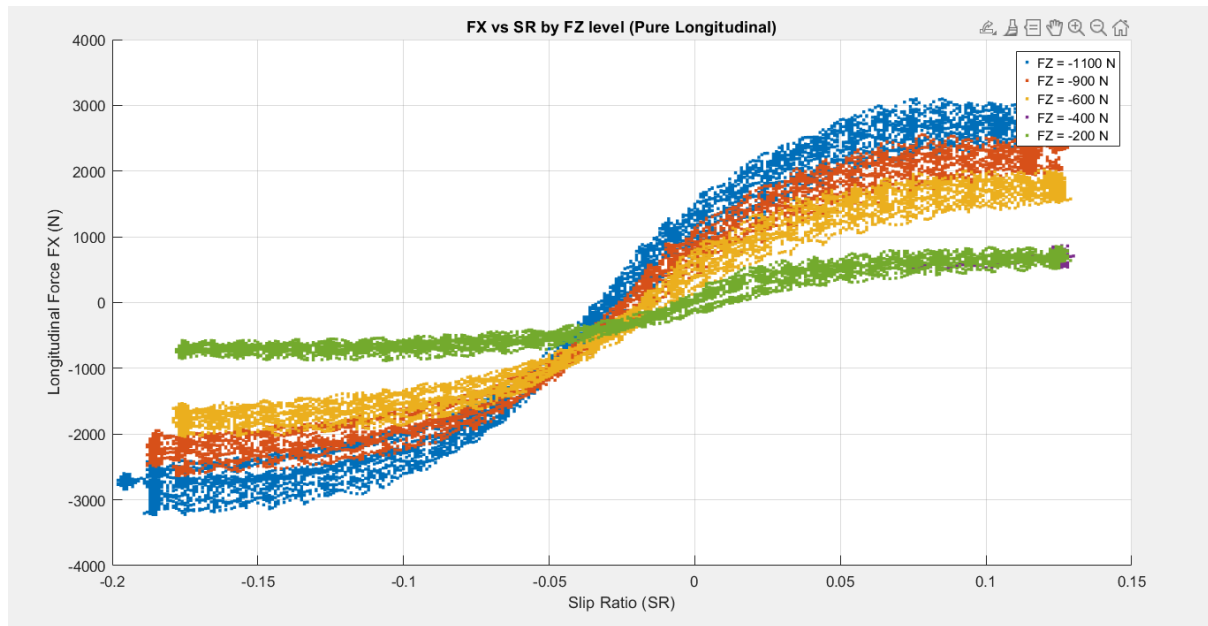
https://drive.google.com/file/d/1I-ND-Nj957_uzoXOCVY0tPA3n46CBPOn/view?usp=sharing



Pure Longitudinal 18"

Dataset used:

<https://drive.google.com/file/d/15zEcfnIBqRw3J55ajHpJAgy9yd1hVk8O/view?usp=sharing>



Now, once we have obtained the graphs for the following datasets, our next task would be to fit the tire model using the Pacejka Magic Formula 5.2 and validate the fitted data (the data predicted by the fitted model) and compare it against the actual data.

Our aim was to obtain a minimum R^2 value of 0.97 and to have the RMSE value as well to be as small as possible. The minimum the RMSE value is and the closer R^2 is to 1, the better is the fitted model.

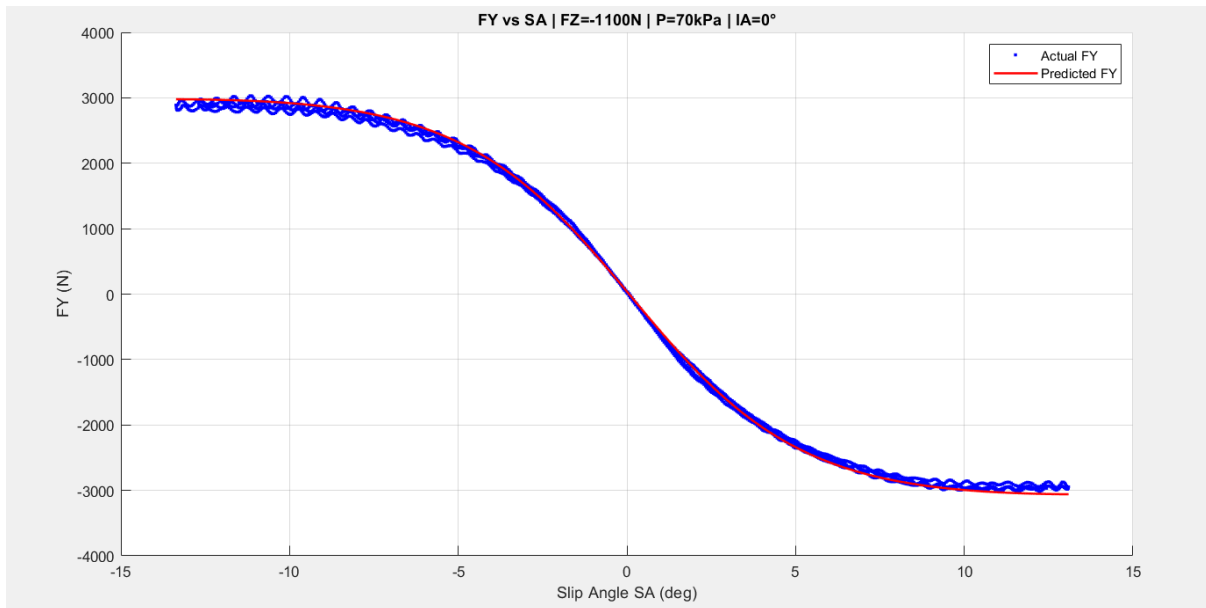
Model Fit and Validation

We used the Pacejka Magic Formula 5.2 to fit the tire model.

- **Model Structure** The MF 5.2 pure longitudinal formula was parameterised by 14 coefficients: $[pC \times 1, pD \times 1 \dots pV \times 1]$, which control the shape (C), peak (D), stiffness (B), curvature (E), and horizontal/vertical shifts (SH, SV) of the F_x curve.
- **Normalised Load (dfz)** Rather than fitting raw F_z , all load-sensitive terms are expressed through $dfz = (F_z - 600) / 600$, so a single set of 14 coefficients captures behaviour continuously across the entire load range simultaneously.
- **Nonlinear Least Squares Optimisation** `lsqcurvefit` minimised the sum of squared residuals between the model output and the generated F_x data across all 5 load cases stacked together, meaning the optimiser found one global parameter set that best fits all loads at once, not separately per load.

<https://in.mathworks.com/help/sdl/ref/tireroadinteractionmagicformula.html>

Pure Cornering 18"

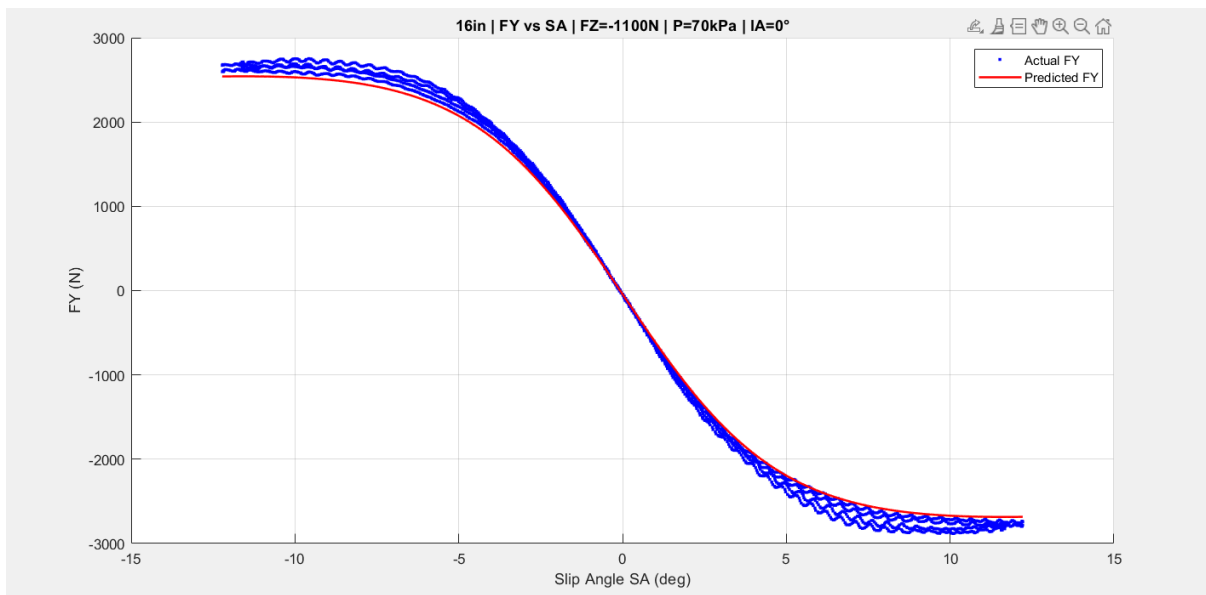


The following plot was obtained for Fz = -1100N and a pressure of 70KPa and 0deg camber.

We obtain RMSE value of 71.26N and R² value of 0.9991 which is a fairly good fit and that could be visualised from the graph as well.

[Pure Cornering 18" – Fitted Function](#)

Pure Cornering 16"

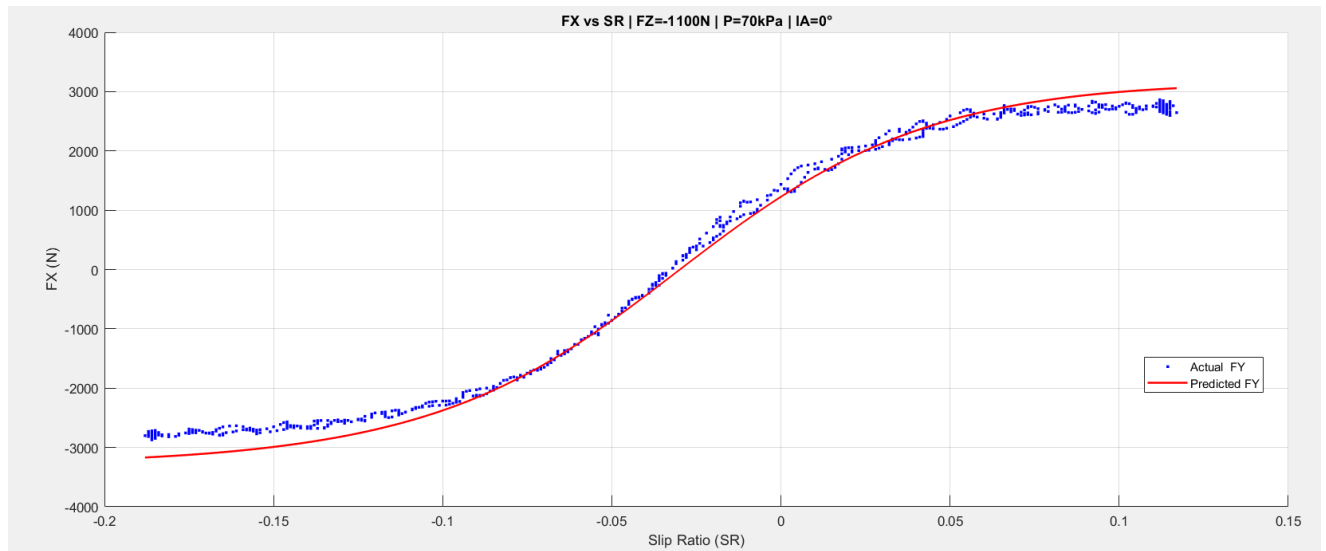


The following plot was obtained for Fz = -1100N and a pressure of 70KPa and 0deg camber.

We obtain RMSE value of 122.29N and R² value of 0.9970 which is a fairly good fit and that could be visualised from the graph as well.

[Pure Cornering 16" – Fitted Function](#)

Pure Longitudinal 18"



The following plot was obtained for $F_z = -1100\text{N}$ and a pressure of 70KPa and 0° camber.

We obtain RMSE value of 235.39N and R^2 value of 0.9893 , although the values are not good as for cornering data but still it could be considered as a fairly good fit and that could be visualised from the comparison shown as well.

[Pure Longitudinal 18" – Fitted Function](#)

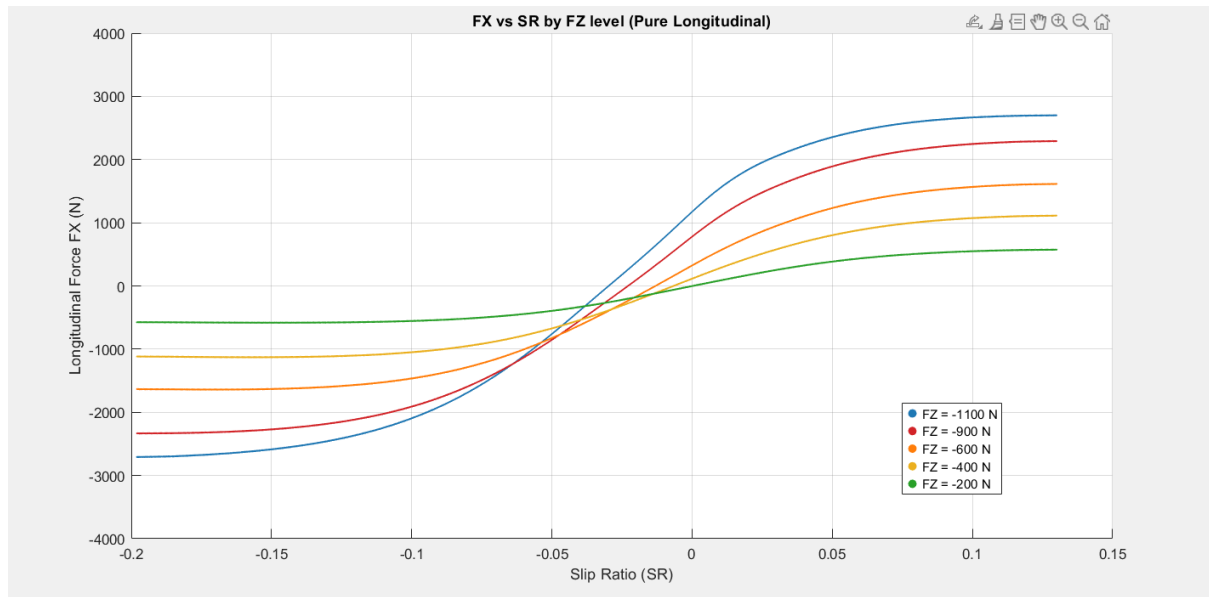
Extrapolating and Validation for Pure Longitudinal 16"

First, we had to make the dataset for pure longitudinal 16" as we did not have the raw TTC datasets for the same. This was done by using a simple philosophy where in we estimated (F_{x_16} as $F_{x_18} * F_{y_16} / F_{y_18}$)

A major challenge we faced while following this approach was in mapping the "slip ratios" with the desired "slip angles". Once pre-processing had been done on the three datasets which we had, in order to use the above relation, we had to carefully map the "slip ratios" with the "slip angles" to get the dataset of F_{x_16} as well.

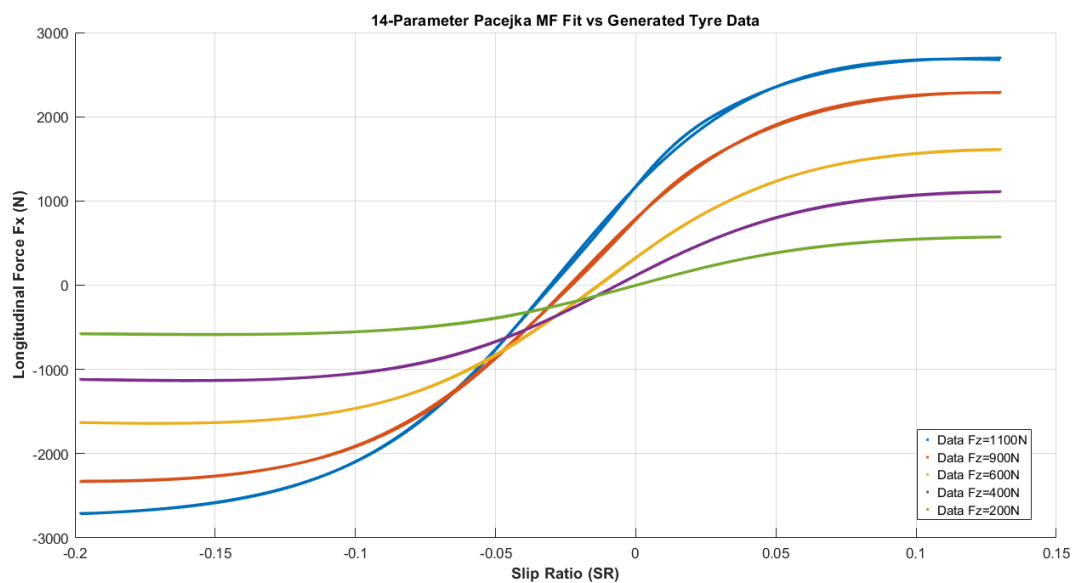
- We divided the SR and SA into 2 clusters each, the first being the sets of positive values for both and the second being the sets of negative values for both.
- Once this was done, we incorporated a simple method to map the values, which was $SA / SA_{\text{max}} = SR / SR_{\text{max}}$ for the set of positive values and $SA / SA_{\text{min}} = SR / SR_{\text{min}}$ for the set of negative values.

Doing this, we were able to map the SR with the corresponding SA values and hence, using the above mentioned formula we obtained the F_{x_16} dataset as well through this simple extrapolation for a particular value of pressure (70KPa).

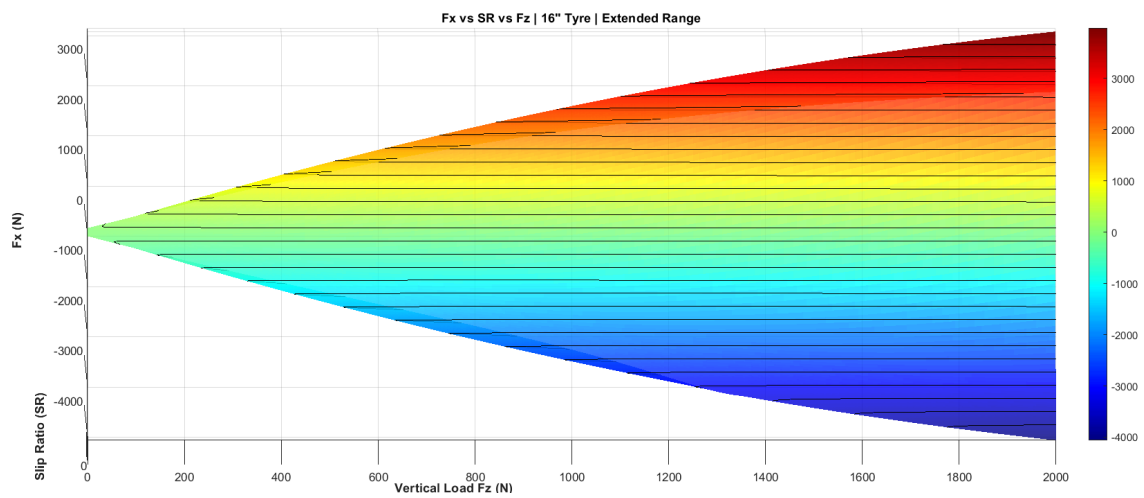
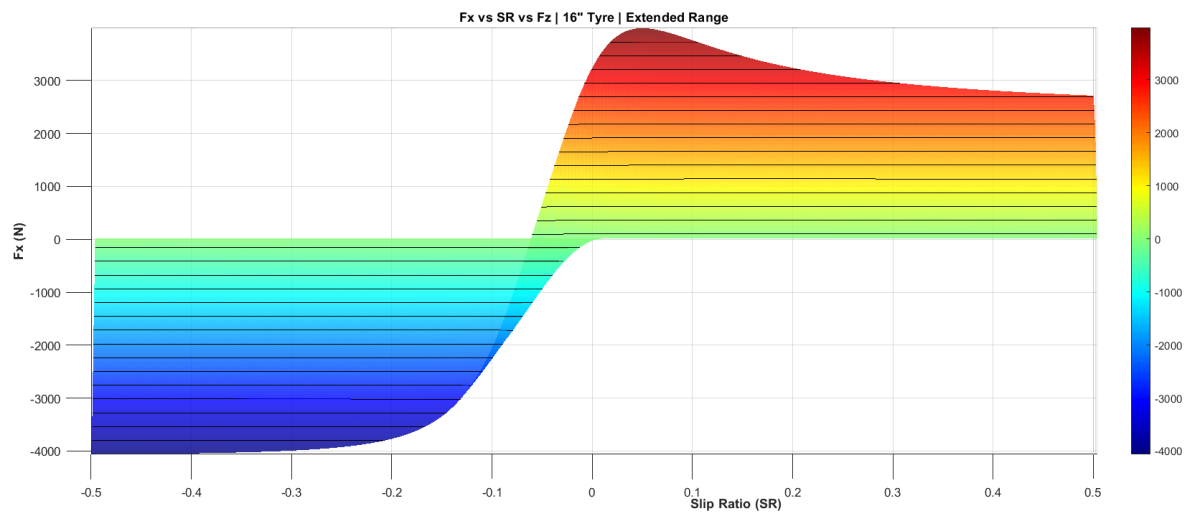
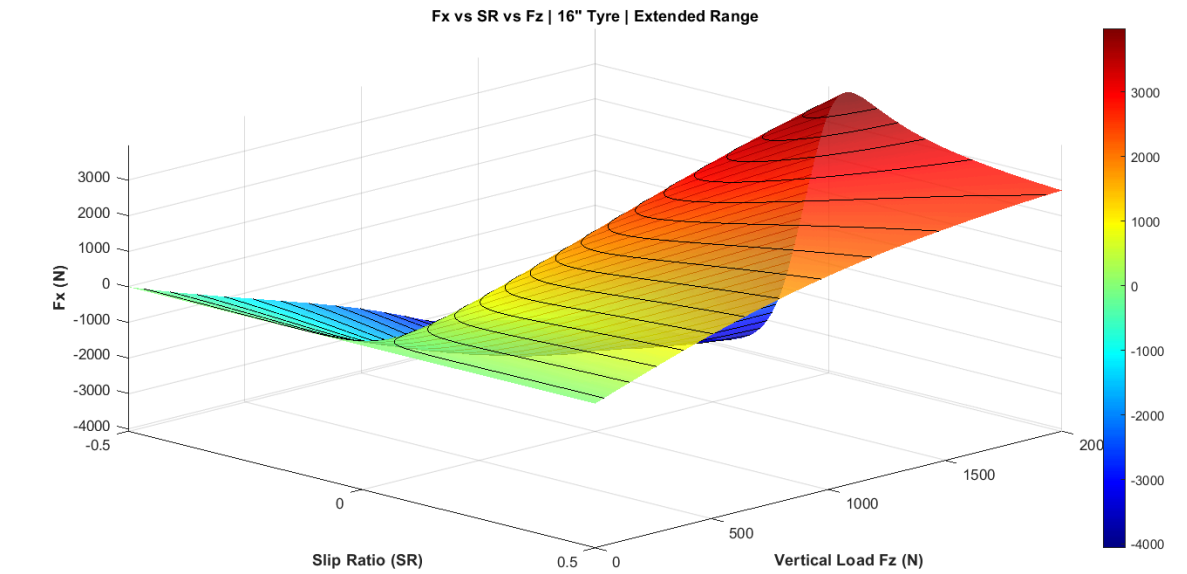


This was the dataset we had generated for Pure Longitudinal 16" tires with the help of existing three datasets and extrapolation.

Now, our next task was to fit a tire model using Pacejka Magic Formula 5.2 and compare between the predicted data of the model and our actual extrapolated data.



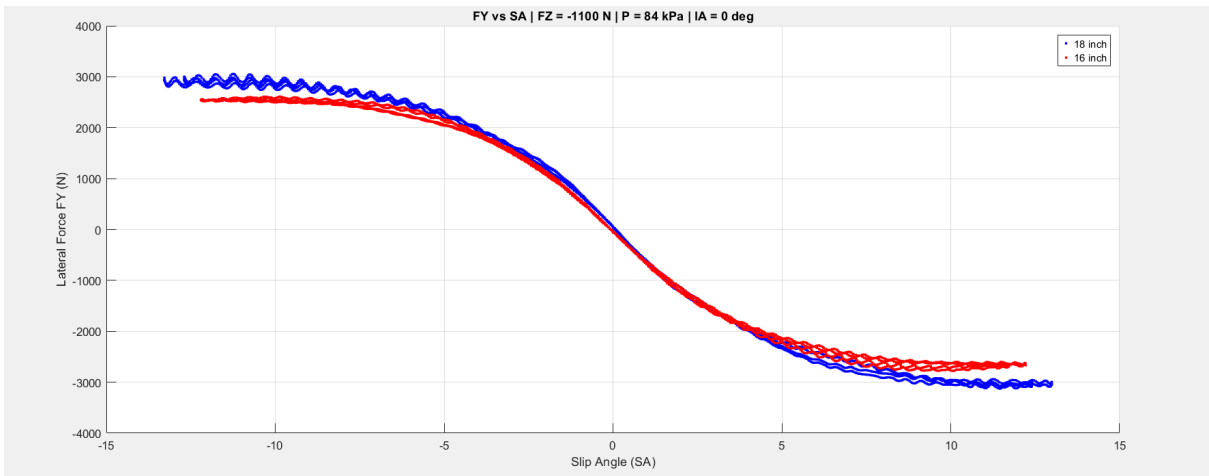
From the data, we can observe that our fitted model almost perfectly overlaps with the extrapolated data, we get the RMSE value to be 23.56 and R^2 value to be 0.9999 for the case of $F_z = -1100$ N and 70 KPa pressure.



[Pure Longitudinal 16" – Fitted Function](#)

Comparison between 16" and 18" Tire Models

Pure Cornering



Pure Longitudinal

